NUS406: INTENSIVE CARE & THEATRE NURSING

Complete Study Guide — Structural Questions & MCQ Ready

Course Instructor: Prof. Eta Vivian Ayamba | Covers all lecture topics + 50+ MCQs with rationale

TABLE OF CONTENTS

SECTION 1: INTRODUCTION TO CRITICAL CARE NURSING

Key Definitions

- **Critical Care**: Specialized medical care for patients with life-threatening conditions requiring continuous monitoring, advanced interventions, and organ system support. Managed in ICU or HDU.
- **Critical Care Nursing**: Specialized nursing focused on critically ill, unstable patients at high risk of life-threatening complications. Uses advanced technology, rapid decision-making, and multidisciplinary collaboration.
- **AACN Definition**: "The specialty that deals specifically with human responses to life-threatening problems."
- **WHO Definition**: Care that improves outcomes of patients with acute life-threatening illnesses through timely and effective interventions.

Characteristics of Critical Care Nursing

- Holistic and patient-centered care
- High level of technical skill
- Evidence-based practice
- Ethical decision-making
- Emotional support for patients and families

Goals of Critical Care Nursing

Preserve life | Restore health | Prevent complications | Provide comfort and support | Assist in end-of-life care when necessary

History of Critical Care Nursing

Era	Key Event
Early Origins (Crimean War)	Florence Nightingale observed critically ill patients needed close observation and grouping together — led to the idea of specialized care areas.
20th Century	Polio epidemic → respiratory failure → mechanical ventilation introduced → specialized monitoring units emerged.
1952 — Birth of ICUs	First modern ICU established in Copenhagen during polio epidemic. Patients grouped for continuous monitoring and mechanical ventilation.
1960s–1970s	Expansion of ICUs worldwide. Development of cardiac care units (CCUs). American Association of Critical-Care Nurses (AACN) founded in 1969.
Modern Era	Advanced life support technologies, specialized training, evidence-based care. Subspecialties: pediatric, neonatal, cardiothoracic critical care.

MCQ TIP: First modern ICU = 1952, Copenhagen, during polio epidemic. AACN founded = 1969. Florence Nightingale = Crimean War origins.

SECTION 2: THE INTENSIVE CARE NURSE

Definition

An ICU nurse is a registered nurse with advanced knowledge/skills in caring for patients with life-threatening conditions, multi-organ dysfunction, and unstable vital signs. Works in ICU, HDU, emergency and trauma units.

Roles of the ICU Nurse

- **Patient Monitoring**: Continuous observation of HR, BP, RR, O2 saturation using advanced equipment.
- **Clinical Care & Interventions**: Administer medications (vasopressors, sedatives), manage mechanical ventilation, perform suctioning, IV line care, catheter management.
- **Rapid Decision-Making**: Recognize early signs of deterioration, respond to emergencies (cardiac arrest), initiate life-saving interventions.
- **Coordination of Care**: Work with multidisciplinary team: doctors, pharmacists, respiratory therapists. Ensure continuity of care.
- **Patient & Family Support**: Emotional/psychological support, educate families, assist with end-of-life decisions.
- **Infection Prevention & Control**: Maintain strict aseptic techniques, prevent HAIs, monitor for infection signs.

Responsibilities

Accurate documentation | Safe use of medical equipment | Maintaining patient safety | Advocacy for patient needs | Ethical and legal accountability

Skills Required

Skill Type	Examples
Technical	Operation of ventilators and monitors; interpretation of ECG and lab results; administration of critical medications
Cognitive	Critical thinking; clinical judgment; problem-solving
Interpersonal	Communication; teamwork; compassion and empathy
Emotional Resilience	Coping with stress; managing grief and loss; maintaining professionalism under pressure

Qualities of a Good ICU Nurse

Vigilance and attention to detail | Quick responsiveness | Strong ethical values | Patience and dedication | Lifelong learner attitude

Challenges Faced by ICU Nurses

- High workload and stress
- Emotional strain from critically ill patients
- Risk of burnout
- Exposure to infectious diseases
- Ethical dilemmas (e.g., end-of-life care)

MCQ TIP: ICU nurse-to-patient ratio = 1:1 or 1:2. Educational requirement = RN + specialized critical care training + AACN certification.

SECTION 3: THE CRITICALLY ILL CLIENT

Definition

A critically ill client is an individual with actual or potential life-threatening organ dysfunction requiring intensive monitoring and support. WHO: severe health conditions needing immediate intervention to sustain life.

Characteristics

- **Physiological Instability**: Unstable vital signs, may need life-support, poor organ function
- **High Risk of Deterioration**: Condition can worsen rapidly; needs constant observation
- **Technology Dependence**: Ventilators, cardiac monitors, infusion pumps, dialysis machines
- **Altered LOC**: Unconscious, sedated, or confused; reduced ability to communicate
- **Complex Medical Needs**: Multiple IV medications, frequent diagnostic tests, multidisciplinary care
- **Impaired Mobility**: Bedridden; risk of pressure ulcers, muscle wasting, blood clots
- **Psychological/Emotional Stress**: Fear, anxiety, confusion, delirium; family members also stressed

Conditions Leading to Critical Illness

Severe infections (sepsis) | Respiratory failure | Cardiac arrest | Trauma (accidents, injuries) | Stroke | Multi-organ failure

Physiological Changes by System

System	Changes
Respiratory	Difficulty breathing, reduced oxygenation, need for O2 therapy or mechanical ventilation
Cardiovascular	Hypotension or hypertension, irregular heart rhythms, poor tissue perfusion
Neurological	Confusion or coma, reduced responsiveness, increased intracranial pressure
Renal	Reduced urine output, fluid and electrolyte imbalance
Gastrointestinal	Reduced digestion, risk of bleeding or ulcers

Patient Needs in Critical Illness

Need Type	Details
Physical	Airway support; adequate oxygenation; fluid/electrolyte balance; nutrition; hygiene; prevent complications
Psychological	Reassurance; reduce fear/anxiety; orientation (for confused patients)
Social	Family interaction; maintaining dignity and respect
Spiritual	Support based on beliefs; access to spiritual care
Safety	Protection from harm and infection; safe environment

Family Needs in Critical Care

Need	Details
Information	Clear, honest, regular updates about patient condition; explanation of procedures

Emotional Support	Counseling, reassurance, help coping with fear and uncertainty
Involvement in Care	Participation in decision-making; opportunity to visit and support patient
Comfort	Waiting areas, rest spaces, access to food and water
Communication	Treated with empathy; opportunity to ask questions

MCQ TIP: ABCDE approach in primary assessment: A=Airway, B=Breathing, C=Circulation, D=Disability (neuro), E=Exposure. Family needs are a distinct exam topic.

SECTION 4: THE ICU — FEATURES, ENVIRONMENT & QUALITY OF CARE

Definition of ICU

A specialized hospital unit designed to provide continuous, intensive care for critically ill patients.

Key Features of the ICU

Feature	Details
Advanced Monitoring	Continuous monitoring of heart rate, BP, oxygen levels; immediate detection of changes
Specialized Equipment	Mechanical ventilators, cardiac monitors, dialysis machines, infusion pumps
Highly Skilled Staff	Specialized nurses, intensivists (doctors), pharmacists, respiratory therapists
Low Nurse-Patient Ratio	1 nurse to 1-2 patients for close observation
24-Hour Care	Continuous care and supervision; immediate emergency response

ICU Environment

Quiet but highly controlled | Bright lights and machine alarms | Strict infection control | Patients admitted when needing intensive monitoring/life support; discharged when condition stabilizes

Qualities of Care for Critically Ill Clients

Quality	Meaning
Holistic Care	Address physical, emotional, psychological, and spiritual needs; treat whole person
Patient-Centered Care	Respect dignity, values, and preferences; involve family when appropriate
Continuous Monitoring	Frequent vital signs assessment; early detection of complications
Timely Intervention	Quick response to emergencies; proper medication/treatment administration
Effective Communication	Clear communication within healthcare team and with patient/family
Safety & Infection Control	Hygiene, sterile techniques; prevent HAIs
Competence & Skill	Well-trained providers; ability to use advanced equipment
Compassion & Empathy	Emotional support; kindness and understanding
Ethical & Legal Responsibility	Confidentiality, patient rights, informed consent
Teamwork & Collaboration	Coordination among all healthcare providers; shared decision-making

SECTION 5a: RESPIRATORY PROBLEMS

Problem	Description	Management
Respiratory Failure	Inability to maintain adequate oxygenation or ventilation	Oxygen therapy or mechanical ventilation
ARDS (Acute Respiratory Distress Syndrome)	Severe lung inflammation; reduced oxygen exchange	Mechanical ventilation; prone positioning; lung-protective strategies
Airway Obstruction	Caused by secretions, swelling, or foreign bodies	Suctioning; airway adjuncts; emergency intubation if needed
Ventilator-Associated Complications	Pneumonia (VAP); lung injury	Prevention bundle: HOB elevation, oral care, daily sedation vacation

SECTION 5b: CARDIOVASCULAR PROBLEMS — SHOCK

Definition of Shock

Shock is a life-threatening emergency where the body fails to deliver enough oxygen to tissues and organs. State of widespread reduced tissue perfusion. Without rapid treatment → irreversible organ damage or death.

Types of Shock — Quick Comparison

Type	Cause / Key Idea	Skin	Pulse	Key Clue
Hypovolemic	Loss of blood/fluid — "Tank is empty"	Cold, clammy	Fast, weak	Bleeding, burns, dehydration
Cardiogenic	Heart pump failure — "Pump broken"	Cold, clammy	Weak/irregular	Heart attack; fluid in lungs
Septic	Severe infection — "Vessels dilate/leak"	Warm → cold	Fast	Known infection (fever)
Anaphylactic	Severe allergy — "Airway + vessels affected"	Flushed, hives	Fast	Minutes after allergen exposure
Neurogenic	Spinal cord injury — "Loss of nerve control"	Warm, dry	SLOW (key!)	Recent spinal trauma
Obstructive	Blocked blood flow (PE, cardiac tamponade)	Pale	Fast	Distended neck veins

MCQ TIP: Neurogenic shock is the ONLY type with SLOW pulse (bradycardia) + warm dry skin. Septic shock starts with WARM skin (early). Anaphylaxis = sudden onset after allergen. All other shocks = cold, clammy skin.

Shock Symptoms to Watch For

- Low blood pressure (weak pulse)
- Cold, clammy, pale skin
- Rapid but weak pulse

- Rapid breathing or difficulty breathing
- Confusion, anxiety, or loss of consciousness
- Low or no urine output

Emergency First Aid for Shock

1. Call emergency services immediately (dial 112 in Cameroon).
2. Lay person flat and elevate legs (unless injury prevents this).
3. Check airway, breathing, circulation; start CPR if needed.
4. Keep them warm with blanket; loosen tight clothing.
5. Do NOT give food or drink.
6. Turn on their side if vomiting (unless spinal injury suspected).

Arrhythmias

Arrhythmia (dysrhythmia): irregular heartbeat due to faulty electrical signals. Normal HR = 60–100 bpm. Tachycardia >100 bpm; Bradycardia <60 bpm.

Type	Description	Risk
Tachycardia	HR >100 bpm	Palpitations, dizziness, stroke risk
Bradycardia	HR <60 bpm	Fainting, fatigue
Atrial Fibrillation (AFib)	Chaotic atrial signals	Major stroke risk
Ventricular Tachycardia	Fast ventricular rhythm	Life-threatening
Ventricular Fibrillation	Chaotic ventricular signals	IMMEDIATE cardiac arrest — most dangerous
PACs/PVCs	Extra beats	Usually harmless; can trigger arrhythmias

MCQ TIP: Ventricular fibrillation = most dangerous arrhythmia → immediate cardiac arrest → CPR + defibrillation. AFib = major stroke risk. Harmless arrhythmias = PACs, PVCs (occasional), sinus arrhythmia.

SECTION 5c: NEUROLOGICAL PROBLEMS

Common Neurological Problems

- Altered level of consciousness (LOC)
- Delirium
- Seizures / Status epilepticus
- Stroke (ischemic or hemorrhagic)
- Traumatic brain injury (TBI)
- Increased intracranial pressure (ICP)
- Hypoxic-ischemic encephalopathy
- Neuromuscular disorders (e.g., critical illness polyneuropathy)

Glasgow Coma Scale (GCS)

Component	Score	Response
Eye Opening (E)	4	Spontaneous
	3	To voice
	2	To pain
	1	None
Verbal Response (V)	5	Oriented
	4	Confused
	3	Inappropriate words
	2	Incomprehensible sounds
	1	None
Motor Response (M)	6	Obeys commands
	5	Localizes pain
	4	Withdraws from pain
	3	Flexion (decorticate)
	2	Extension (decerebrate)
	1	None

Max GCS = 15 (fully conscious). Min = 3 (deep coma). GCS ≤8 = severe; consider intubation.

Increased ICP — Cushing's Triad (CRITICAL MCQ)

Cushing's Triad = Hypertension + Bradycardia + Irregular respiration. ICP >20 mmHg is pathological. Causes: TBI, hemorrhage, tumors, cerebral edema. Management: Head elevation 30°, sedation, osmotic agents (mannitol), controlled ventilation, surgical decompression.

Condition	Key Causes	Diagnosis Tool	Management

Altered LOC	Stroke, metabolic, sepsis, toxins	GCS, labs, CT/MRI, EEG	ABCs, correct reversible causes
Delirium	Sepsis, hypoxia, drugs, ICU environment	CAM-ICU, ICDSC	Non-pharmacological bundle; treat cause
Raised ICP	TBI, hemorrhage, stroke, tumors	Clinical signs, CT/MRI, ICP monitoring	Osmotherapy, hyperventilation, surgery

MCQ TIP: Cushing's triad = hypertension + bradycardia + irregular breathing (sign of raised ICP). Delirium screening in ICU uses CAM-ICU. Decorticate = arms flex; decerebrate = arms extend (worse sign).

SECTION 5d: RENAL PROBLEMS

Acute Kidney Injury (AKI)

KDIGO Criteria: Serum creatinine ↑ ≥0.3 mg/dL within 48h, OR ≥1.5× baseline within 7 days, OR urine output <0.5 mL/kg/h for ≥6h.

Type	Causes
Pre-renal	Hypovolemia, hypotension, sepsis, shock, renal hypoperfusion
Intrinsic Renal	Acute tubular necrosis (ischemia, toxins), glomerulonephritis, interstitial nephritis
Post-renal	Obstruction (stones, tumors, catheter blockage)

Management of AKI

- Optimize hemodynamics (fluids, vasopressors)
- Avoid nephrotoxins (NSAIDs, aminoglycosides, contrast)
- Adjust drug dosing to renal function
- Renal replacement therapy (RRT) indications: severe acidosis, hyperkalemia, fluid overload, uremic complications
- RRT modalities: intermittent hemodialysis or continuous RRT (CRRT)

Fluid & Electrolyte Imbalances

Electrolyte	Imbalance	Cause	Management
Sodium	Hyponatremia	SIADH, fluid overload	Slow correction with hypertonic saline if symptomatic
Sodium	Hypernatremia	Dehydration, diabetes insipidus	Gradual fluid replacement
Potassium	Hyperkalemia	AKI, tissue breakdown	Calcium gluconate, insulin+glucose, dialysis if severe
Potassium	Hypokalemia	Diuretics, GI losses	IV/oral potassium replacement
Calcium	Hypocalcemia	Sepsis, pancreatitis	IV calcium replacement

MCQ TIP: AKI affects 30-60% of ICU patients. Hyperkalemia = most life-threatening electrolyte disorder (causes arrhythmias). Nephrotoxins to avoid: NSAIDs, aminoglycosides, contrast dye.

SECTION 5e: GASTROINTESTINAL PROBLEMS

Problem	Causes	Diagnosis	Management
Malnutrition	Critical illness, prolonged fasting, malabsorption, increased metabolic demand	Weight loss, low albumin; screening tools: NRS-2002, MUST	Early enteral nutrition (within 24-48h); parenteral if enteral not possible; monitor for refeeding syndrome
Stress Ulcers	Severe illness, mechanical ventilation >48h, coagulopathy, mucosal ischemia	Hematemesis, melena, anemia; endoscopy confirms	Prophylaxis: PPIs or H2 blockers in high-risk patients; endoscopic hemostasis if bleeding
Constipation	Opioids, immobility, dehydration, reduced gut motility	Infrequent stools, abdominal distension	Laxatives (stimulant/osmotic); hydration; early mobilization; minimize opioids
Diarrhea	Enteral feeding intolerance, C. difficile, antibiotics	Frequent loose stools, dehydration, electrolyte imbalance	Identify cause; hydration; adjust enteral feeding; infection control

MCQ TIP: Enteral nutrition is PREFERRED over parenteral (if feasible). Stress ulcer prophylaxis = PPIs or H2 blockers. Refeeding syndrome = hypophosphatemia + hypokalemia + hypomagnesemia.

SECTION 5f: INFECTION PROBLEMS

Sepsis

Life-threatening organ dysfunction caused by dysregulated host response to infection. Septic shock = subset with profound circulatory, cellular, and metabolic abnormalities.

Screening tools: qSOFA (hypotension + altered mentation + tachypnea); SOFA score (organ dysfunction).

Sepsis Bundle (within 1 hour): Obtain cultures → Administer antibiotics → IV fluid resuscitation (30 mL/kg crystalloid) → Vasopressors (norepinephrine) if hypotension persists.

Hospital-Acquired Infections (HAIs)

Infections not present at admission, occurring ≥48 hours after hospitalization.

Type	Risk Factor	Prevention
VAP (Ventilator-Associated Pneumonia)	Endotracheal intubation	HOB elevation 30-45°, oral care, daily sedation vacation, early weaning
CAUTI (Catheter-Associated UTI)	Urinary catheters	Remove early; aseptic insertion; closed drainage system
CLABSI (Central Line Bloodstream Infection)	Central venous catheters	Sterile insertion technique; daily review for removal need
SSI (Surgical Site Infection)	Surgery	Pre-op antibiotics; sterile technique; wound care

MCQ TIP: HAIs occur ≥48h after admission. Most common HAI in ICU = VAP. Norepinephrine = first-line vasopressor for septic shock. Sepsis bundle = within 1 HOUR.

SECTION 5g: OTHER PROBLEMS

Problem	Causes	Management
Pressure Ulcers (Bedsores)	Prolonged immobility, poor perfusion	Reposition every 2h; pressure-relieving devices; skin hygiene
Muscle Wasting	Immobility, critical illness	Early passive/active exercises; physiotherapy
DVT (Deep Vein Thrombosis)	Immobility, blood clots	Prophylactic anticoagulation; compression stockings; early mobilization
ICU Delirium	Noise, sleep deprivation, medications, sepsis	CAM-ICU screening; reorientation; minimize sedation; family visits
Anxiety/Depression	ICU environment, illness, uncertainty	Reassurance; counseling; involve family
Communication Problems	Intubation (unable to speak), weakness	Writing boards; picture boards; lip reading; alternative communication
Sleep Disturbances	Noise, lights, frequent procedures	Cluster care activities; reduce noise; dim lights at night
Pain	Illness, procedures, immobility	Regular pain assessment; analgesics; comfort positioning

SECTION 6: MONITORING THE CRITICALLY ILL CLIENT

Purpose of Monitoring

Detect early signs of deterioration | Guide treatment | Evaluate response to therapy | Maintain vital functions | Prevent complications

Type	What it Assesses	Tools/Methods
Clinical (Physical)	LOC, skin color/temp, capillary refill, urine output, pain	GCS, direct observation
Vital Signs	Temperature, pulse, BP, respiratory rate	Thermometer, BP cuff, monitors
Hemodynamic	Cardiovascular function and blood flow	Non-invasive: BP cuff, pulse oximeter; Invasive: arterial line, CVP
Respiratory	Oxygenation, ventilation	SpO2 (pulse oximeter), ABG, ventilator parameters
Cardiac	Heart rhythm and rate	Continuous ECG monitoring
Neurological	Consciousness, ICP signs	GCS, pupillary response, Cushing's triad monitoring
Renal	Kidney function, fluid balance	Urine output charting, urea/creatinine blood tests
Laboratory	Blood chemistry, infection markers	Electrolytes, Hb, WBC, glucose, ABG

Nursing Responsibilities in Monitoring

- Perform frequent and accurate assessments
- Record vital signs and document changes accurately
- Understand normal vs abnormal values; recognize trends
- Report abnormal findings immediately to healthcare team
- Ensure proper functioning and hygiene of monitoring devices

Principles of Effective Monitoring

Continuous and consistent | Accurate and timely | Patient-centered | Use both technology AND clinical judgment | Early intervention based on findings

Complications of Monitoring

Infection (especially invasive monitoring) | Equipment malfunction | Misinterpretation of data | Patient discomfort

SECTION 7: NURSING CARE PLAN OF THE CRITICALLY ILL CLIENT

Goals of Nursing Care

Maintain ABC | Stabilize vital functions | Prevent complications | Provide comfort/pain relief | Support psychological well-being | Promote recovery and rehabilitation

Key Nursing Interventions by Domain

Domain	Interventions
Airway & Breathing	Maintain patency (position, suction); oxygen therapy; assist ventilation; monitor SpO2
Circulatory Support	Monitor vital signs; administer IV fluids/medications; observe for shock signs
Fluid & Electrolytes	Monitor I&O; assess for dehydration/overload; administer fluids as prescribed
Pain Management	Assess pain regularly; administer analgesics; positioning; calm environment
Nutrition	Enteral or parenteral feeding; monitor nutritional status; prevent aspiration
Infection Control	Hand hygiene; aseptic technique; wound/IV line care; monitor for infection
Skin & Pressure Ulcer	Reposition every 2 hours; skin hygiene; pressure-relieving devices
Mobility	Passive/active exercises; prevent DVT, muscle wasting
Neurological	Monitor GCS; watch for raised ICP signs; maintain patient safety
Psychological	Reassurance; reduce anxiety; orient confused patients
Communication	Clear communication; alternative methods (writing boards) for intubated patients
Family Care	Regular updates; emotional support; involve in care; respect cultural beliefs

Nursing Care Plan — Summary Table

Nursing Diagnosis	Goal	Interventions	Rationale
Impaired Gas Exchange r/t respiratory failure	Maintain SpO2 >94%	Monitor RR & SpO2; O2 therapy; semi-Fowler's position; assist ventilation	Detects hypoxia; improves O2 delivery; enhances lung expansion
Decreased Cardiac Output r/t circulatory failure	Stable vital signs	Monitor BP, pulse, ECG; IV fluids/meds; assess for shock	Evaluates heart function; maintains perfusion
Risk for Infection r/t invasive procedures	No infection	Hand hygiene; aseptic technique; monitor temp & WBC; IV/catheter care	Prevents microorganism spread; early detection
Acute Pain r/t illness/procedures	Reduced pain	Assess pain; administer analgesics; comfort measures	Ensures pain control; promotes relaxation
Imbalanced Nutrition: Less than Body Req.	Adequate nutrition	Enteral/parenteral feeding; monitor weight/labs; prevent aspiration	Adequate nutrients; supports healing
Impaired Skin Integrity r/t immobility	Intact skin	Reposition q2h; skin hygiene; pressure devices	Prevents pressure ulcers; promotes circulation

Risk for Fluid Volume Imbalance	Normal fluid balance	Monitor I&O; assess edema/dehydration; fluids as prescribed	Detects imbalance; maintains homeostasis
Anxiety r/t critical condition	Reduced anxiety	Emotional support; clear information; family involvement	Reduces fear; builds trust
Impaired Physical Mobility	Optimal mobility	Passive/active exercises; reposition; prevent DVT	Prevents immobility complications

SECTION 8: MCQ TIPS & PRACTICE QUESTIONS

Top MCQ Traps — NUS406

- **First modern ICU**: 1952, Copenhagen, Denmark, during the POLIO epidemic — not World War I or II.
- **AACN Founded**: 1969 — American Association of Critical-Care Nurses.
- **Neurogenic shock pulse**: SLOW (bradycardia) + warm dry skin — opposite of all other shocks.
- **Septic shock early vs late**: Early septic shock = WARM, flushed skin. Late = cold, clammy (like other shocks).
- **Ventricular fibrillation**: MOST dangerous arrhythmia → cardiac arrest → requires IMMEDIATE CPR + defibrillation.
- **Cushing's triad**: Hypertension + Bradycardia + Irregular respiration = sign of RAISED ICP (not cardiac shock).
- **GCS maximum score**: 15 (normal). Minimum = 3 (deep coma). ≤8 = severe; usually intubate.
- **HAI definition**: Infection NOT present at admission; occurs ≥48 HOURS after hospitalization.
- **Sepsis bundle timing**: WITHIN 1 HOUR — obtain cultures, antibiotics, fluids, vasopressors if needed.
- **AKI criteria**: Creatinine ↑ ≥0.3 mg/dL in 48h OR urine output <0.5 mL/kg/h for ≥6h.
- **Enteral vs parenteral**: Enteral nutrition = PREFERRED if gut works. Parenteral = only when enteral not possible.
- **Stress ulcer prophylaxis**: PPIs or H2 blockers — prevention is key, especially for ventilated patients >48h.
- **ICU nurse-patient ratio**: 1:1 or 1:2 — this is a defining feature of ICU care.
- **ABCDE primary assessment**: A=Airway, B=Breathing, C=Circulation, D=Disability (neuro), E=Exposure.
- **Refeeding syndrome electrolytes**: Hypophosphatemia + hypokalemia + hypomagnesemia after refeeding a malnourished patient.

50 Practice MCQs with Answers & Rationale

1. Critical care nursing is defined by the AACN as:

A. Nursing care for elderly patients only.
B. The specialty that deals with human responses to life-threatening problems. ✓
C. Nursing care provided only in emergency departments.
D. General nursing care for all hospital patients.

Rationale: The AACN defines critical care nursing as the specialty dealing specifically with human responses to life-threatening problems.

2. The first modern ICU was established in:

A. 1940, London
B. 1945, New York
C. 1952, Copenhagen ✓
D. 1969, Washington D.C.

Rationale: The first modern ICU was established in 1952 in Copenhagen, Denmark, during the polio epidemic, when patients needed ventilatory support.

3. Florence Nightingale's contribution to critical care was:

A. Inventing the ventilator
B. Founding the AACN
C. Observing that critically ill patients needed close monitoring and grouping ✓
D. Developing the ICU admission criteria

Rationale: During the Crimean War, Nightingale observed that critically ill patients needed close observation and grouping — laying the groundwork for specialized care areas.

4. The American Association of Critical-Care Nurses was founded in:

A. 1952
B. 1960
C. 1969 ✓
D. 1975

Rationale: The AACN was founded in 1969 during the expansion of ICUs worldwide in the 1960s–1970s.

5. Which of the following is NOT a goal of critical care nursing?

A. Preserve life
B. Restore health
C. Perform surgery ✓
D. Assist in end-of-life care

Rationale: Goals of critical care nursing include preserving life, restoring health, preventing complications, providing comfort, and assisting in end-of-life care. Surgery is performed by surgeons, not nurses.

6. The typical nurse-to-patient ratio in the ICU is:

A. 1:5
B. 1:3
C. 1:1 to 1:2 ✓
D. 1:10

Rationale: ICUs maintain a low nurse-patient ratio (1:1 or 1:2) for close observation, which is a defining feature of intensive care.

7. Which skill involves the ability to interpret ECG results and operate ventilators?

A. Cognitive skills
B. Technical skills ✓
C. Interpersonal skills
D. Emotional skills

Rationale: Technical skills in ICU nursing include operating ventilators and monitors, interpreting ECG and lab results, and administering critical medications.

8. A critically ill client is BEST defined as:

A. Any patient admitted to hospital
B. A patient with mild illness needing observation
C. A patient with actual or potential life-threatening organ dysfunction requiring intensive monitoring ✓
D. A patient recovering from surgery

Rationale: A critically ill client has actual or potential life-threatening organ dysfunction requiring intensive monitoring, advanced support, and specialized care.

9. Which is a characteristic specific to critically ill clients?

A. Stable vital signs
B. Independence from medical technology
C. Dependence on life-support systems like ventilators ✓
D. Full ability to communicate needs

Rationale: Critically ill clients characteristically depend on life-support systems such as ventilators, cardiac monitors, and infusion pumps.

10. The ABCDE approach to primary assessment stands for:

A. Airway, Blood, Chemistry, Diagnosis, Evaluation
B. Airway, Breathing, Circulation, Disability, Exposure ✓
C. Assessment, Breathing, Circulation, Drugs, Examination
D. Airway, Bones, Circulation, Documentation, Evaluation

Rationale: ABCDE = Airway (ensure clear), Breathing (assess effort), Circulation (pulse/BP), Disability (neurological), Exposure (full body exam).

11. Family needs in critical care include all of the following EXCEPT:

A. Information about patient condition
B. Emotional support and counseling
C. Permission to perform nursing procedures ✓
D. Involvement in care and decision-making

Rationale: Family needs include information, emotional support, involvement in care, comfort, and respectful communication — not permission to perform nursing procedures.

12. Which type of shock is characterized by a SLOW pulse and warm, dry skin?

A. Hypovolemic
B. Cardiogenic
C. Neurogenic ✓
D. Septic

Rationale: Neurogenic shock (from spinal cord/brain injury) causes bradycardia (slow pulse) + warm, dry skin due to loss of sympathetic nerve control — the key distinguishing feature.

13. A patient is brought in after a road traffic accident with low BP and cold, clammy skin. Which type of shock is most likely?

A. Neurogenic
B. Anaphylactic
C. Hypovolemic ✓
D. Septic

Rationale: Hypovolemic shock from trauma (blood loss) presents with low BP, cold/clammy skin, rapid weak pulse — the 'empty tank' scenario.

14. Septic shock in its early stage is characterized by:

A. Cold, clammy skin
B. Warm, flushed skin with low BP ✓
C. High BP
D. Slow pulse

Rationale: Early septic shock presents with warm, flushed skin (vasodilation from infection) + low BP. Late stages progress to cold, clammy skin.

15. Anaphylactic shock occurs:

A. Hours after allergen exposure
B. Days after infection
C. Within minutes of allergen exposure ✓
D. Only in elderly patients

Rationale: Anaphylactic shock typically develops within minutes of exposure to an allergen (food, medication, insect sting).

16. The most dangerous arrhythmia leading to immediate cardiac arrest is:

A. Sinus bradycardia
B. Atrial fibrillation
C. Premature ventricular contractions
D. Ventricular fibrillation ✓

Rationale: Ventricular fibrillation causes chaotic ventricular signals → no effective cardiac output → immediate cardiac arrest. Requires CPR + defibrillation.

17. A patient with a heart rate >100 bpm has:

A. Bradycardia
B. Tachycardia ✓
C. Atrial fibrillation
D. Heart block

Rationale: Tachycardia = heart rate >100 bpm. Bradycardia = <60 bpm. Normal = 60–100 bpm.

18. Which arrhythmia carries the highest risk of stroke?

A. Sinus bradycardia
B. Premature atrial contractions
C. Atrial fibrillation (AFib) ✓
D. Sinus tachycardia

Rationale: Atrial fibrillation causes chaotic atrial signals leading to blood pooling and clot formation in the atria, which can embolize to the brain causing stroke.

19. Cushing's Triad (sign of raised ICP) consists of:

A. Hypotension, tachycardia, shallow breathing
B. Hypertension, bradycardia, irregular respiration ✓
C. Fever, tachycardia, hypotension
D. Low BP, fast pulse, warm skin

Rationale: Cushing's Triad = Hypertension + Bradycardia + Irregular respiration. This is a late, ominous sign of severely raised intracranial pressure.

20. The maximum score on the Glasgow Coma Scale is:

A. 10
B. 12
C. 15 ✓
D. 20

Rationale: GCS max = 15 (E4+V5+M6). Min = 3 (all no response). GCS ≤8 = severe brain injury; consider airway protection.

21. ICP is considered pathological when it exceeds:

A. 5 mmHg
B. 10 mmHg
C. 15 mmHg
D. 20 mmHg ✓

Rationale: Normal ICP = 7–15 mmHg. Sustained ICP >20 mmHg is pathological and can lead to brain herniation and death.

22. Delirium in the ICU is BEST screened using:

A. GCS alone
B. CAM-ICU or ICDSC ✓
C. MRI
D. EEG only

Rationale: CAM-ICU (Confusion Assessment Method for ICU) and ICDSC (Intensive Care Delirium Screening Checklist) are validated screening tools for ICU delirium.

23. Which of the following is a recommended position to reduce ICP?

A. Flat (supine)
B. Trendelenburg (head down)
C. Head of bed elevated at 30° ✓
D. Prone position

Rationale: Elevating the head of bed to 30° promotes venous drainage from the brain, reduces ICP. Trendelenburg would worsen it.

24. Acute Kidney Injury (AKI) is diagnosed when urine output falls below:

A. 1 mL/kg/h for 2h
B. 0.5 mL/kg/h for ≥6h ✓
C. 2 mL/kg/h for 4h
D. 0.1 mL/kg/h for 1h

Rationale: KDIGO AKI criteria include urine output <0.5 mL/kg/h for ≥6h, OR serum creatinine rise ≥0.3 mg/dL in 48h, OR ≥1.5× baseline within 7 days.

25. Which medication is CONTRAINDICATED in AKI due to nephrotoxicity?

A. Paracetamol
B. NSAIDs ✓
C. Antihistamines
D. Antacids

Rationale: NSAIDs (and aminoglycosides, contrast dye) are nephrotoxic and must be AVOIDED in AKI. They reduce renal blood flow by inhibiting prostaglandins.

26. The most life-threatening electrolyte imbalance in AKI is:

A. Hyponatremia
B. Hypocalcemia
C. Hyperkalemia ✓
D. Hypomagnesemia

Rationale: Hyperkalemia is the most immediately life-threatening electrolyte disorder in AKI because it can cause fatal cardiac arrhythmias. Treatment: calcium gluconate, insulin+glucose, dialysis.

27. Enteral nutrition is preferred over parenteral nutrition because:

A. It is cheaper and easier
B. It maintains gut integrity, reduces infection risk, and has better outcomes ✓
C. It requires less monitoring
D. It provides more calories

Rationale: Enteral nutrition (via gut) maintains intestinal barrier integrity, reduces bacterial translocation, and has fewer complications than parenteral. Preferred if gut is functional.

28. Which group of patients should receive stress ulcer prophylaxis?

A. All ICU patients
B. Only post-surgical patients
C. High-risk patients: mechanical ventilation >48h, coagulopathy ✓
D. Only patients with known peptic ulcer disease

Rationale: Stress ulcer prophylaxis is targeted to high-risk ICU patients: mechanical ventilation >48h, coagulopathy, severe illness. PPIs or H2 blockers are used.

29. Refeeding syndrome is characterized by:

A. Hyperkalemia, hyperphosphatemia, hypermagnesemia
B. Hypophosphatemia, hypokalemia, hypomagnesemia ✓
C. Hypernatremia and hyperglycemia
D. High albumin and weight gain

Rationale: Refeeding syndrome occurs when malnourished patients are re-fed too quickly. Insulin release shifts electrolytes into cells → hypophosphatemia (most hallmark), hypokalemia, hypomagnesemia.

30. Hospital-acquired infections (HAIs) are defined as infections occurring:

A. Within 24 hours of admission
B. ≥48 hours after admission, not present at time of admission ✓
C. Only in ICU patients
D. Only from surgical procedures

Rationale: HAIs (nosocomial infections) are infections not present/incubating at admission, developing ≥48 hours after hospitalization.

31. The most common VAP prevention measure is:

A. Prone positioning
B. Head of bed elevation to 30-45° and daily oral care ✓
C. Immediate extubation
D. Broad-spectrum antibiotics prophylactically

Rationale: VAP bundle includes: head of bed elevation (30-45°), daily oral decontamination, daily sedation vacation to assess extubation readiness, subglottic secretion drainage.

32. The sepsis bundle should be completed within:

A. 6 hours
B. 3 hours
C. 1 hour ✓
D. 24 hours

Rationale: Current Surviving Sepsis Campaign recommends completing the sepsis bundle (cultures, antibiotics, fluids, vasopressors) within 1 HOUR of sepsis recognition.

33. First-line vasopressor for septic shock is:

A. Dopamine
B. Adrenaline (epinephrine)
C. Norepinephrine ✓
D. Vasopressin

Rationale: Norepinephrine is the first-line vasopressor for septic shock per international guidelines (Surviving Sepsis Campaign).

34. Which is a primary neurological problem (not secondary/systemic)?

A. Hypoglycemia-induced altered consciousness

B. Sepsis-associated encephalopathy

C. Ischemic stroke ✓

D. Drug overdose causing coma

Rationale: Stroke is a primary neurological cause. The others (hypoglycemia, sepsis, drugs) are secondary/systemic causes of altered consciousness.

35. Repositioning a critically ill patient should be done every:

A. 4 hours

B. 6 hours

C. 2 hours ✓

D. 8 hours

Rationale: Patients should be repositioned every 2 hours to prevent pressure ulcers and promote circulation. This is a standard nursing intervention for immobile patients.

36. Which monitoring type assesses LOC using the Glasgow Coma Scale?

A. Hemodynamic monitoring

B. Respiratory monitoring

C. Neurological monitoring ✓

D. Laboratory monitoring

Rationale: Neurological monitoring includes LOC assessment (GCS), pupil size and reaction, and signs of increased ICP.

37. Central venous pressure (CVP) monitoring is classified as:

A. Clinical monitoring

B. Non-invasive hemodynamic monitoring

C. Invasive hemodynamic monitoring ✓

D. Laboratory monitoring

Rationale: CVP is an invasive hemodynamic monitoring method (requires central venous catheter). Non-invasive = BP cuff, pulse oximeter.

38. An ICU patient cannot speak because they are intubated. The BEST alternative communication method is:

A. Waiting until extubation

B. Asking family to interpret

C. Writing boards or picture boards ✓

D. Lip reading only

Rationale: For intubated patients who cannot speak, alternative communication tools include writing boards, picture/communication boards, eye-gaze systems, and electronic devices.

39. Which nursing diagnosis is MOST appropriate for a patient on mechanical ventilation?

A. Risk for falls

B. Impaired gas exchange related to respiratory failure ✓

C. Constipation related to immobility

D. Risk for social isolation

Rationale: A patient on mechanical ventilation has impaired gas exchange as the priority nursing diagnosis. The ventilator is being used because normal gas exchange is compromised.

40. What is the rationale for positioning a critically ill patient in semi-Fowler's position?

A. Reduces risk of falls

B. Enhances lung expansion and improves oxygenation ✓
C. Prevents DVT
D. Reduces pain

Rationale: Semi-Fowler's position (head of bed 30-45°) allows the diaphragm to descend, enhances lung expansion, improves oxygenation, and reduces aspiration risk.

41. The primary goal of nursing care for the critically ill is:

A. Document all medications administered
B. Maintain airway, breathing, and circulation (ABC) ✓
C. Notify the family of diagnosis
D. Prepare the patient for surgery

Rationale: The primary goal of nursing care for the critically ill is to maintain ABC (Airway, Breathing, Circulation) — the life-sustaining functions.

42. ICU delirium is BEST prevented by:

A. Increasing sedation doses
B. Keeping the patient isolated
C. Reorientation, sleep hygiene, early mobilization, family engagement ✓
D. Routine antipsychotic medications

Rationale: Non-pharmacological bundle for ICU delirium: reorientation (clocks, calendars), sleep hygiene, early mobilization, family visits, minimize sedation. Pharmacological options have limited evidence.

43. Which condition is caused by severe physiological stress and can lead to GI bleeding?

A. Stress ulcers ✓
B. Pressure ulcers
C. Peptic ulcer disease
D. Crohn's disease

Rationale: Stress ulcers are acute mucosal erosions in the stomach/duodenum occurring in critically ill patients due to severe physiological stress, hypoperfusion, and mucosal ischemia.

44. What does qSOFA stand for in sepsis screening?

A. Quick Sequential Organ Failure Assessment ✓
B. Quantitative Sepsis Fever Assessment
C. Quick Sepsis Organ Function Analysis
D. Quantitative Systematic Organ Failure Approach

Rationale: qSOFA = Quick Sequential [Sepsis-related] Organ Failure Assessment. Positive when ≥2 of: altered mentation, tachypnea (RR ≥22), hypotension (SBP ≤100).

45. Deep vein thrombosis (DVT) in critically ill patients is caused by:

A. Excessive hydration
B. Prolonged immobility ✓
C. High oxygen therapy
D. Pain medication

Rationale: Prolonged immobility → venous stasis → blood clot formation in deep veins (DVT). Prevention: prophylactic anticoagulation, compression stockings, early mobilization.

46. A patient with a history of penicillin allergy develops sudden onset swelling of the lips, difficulty breathing, and low BP after receiving amoxicillin. This is:

A. Septic shock

B. Hypovolemic shock

C. Anaphylactic shock ✓

D. Cardiogenic shock

Rationale: Anaphylactic shock = sudden life-threatening allergic reaction. Classic features: rapid onset after allergen exposure, facial swelling, breathing difficulty, hives, low BP.

47. For a patient with raised ICP, which of the following should be AVOIDED?

A. Head elevation at 30°

B. Osmotic diuretics (mannitol)

C. Trendelenburg (head-down) position ✓

D. Sedation

Rationale: Trendelenburg position (head down) increases venous pressure in the head and worsens ICP. All other options help manage raised ICP.

48. An unconscious patient's eyes open only to pain. What is the eye-opening score on GCS?

A. 1

B. 2 ✓

C. 3

D. 4

Rationale: GCS Eye Opening: 4=spontaneous, 3=to voice, 2=to pain, 1=none. Eyes opening only to pain = E2.

49. Which statement BEST describes holistic care in critical illness?

A. Focusing only on the patient's physical needs

B. Treating only the disease, not the person

C. Addressing physical, emotional, psychological, and spiritual needs of the patient ✓

D. Providing care only during emergencies

Rationale: Holistic care addresses ALL dimensions of the person: physical, emotional, psychological, and spiritual — treating the patient as a whole, not just the disease.

50. The primary reason for maintaining strict hand hygiene in the ICU is to:

A. Improve patient comfort

B. Prevent hospital-acquired infections ✓

C. Reduce medication errors

D. Maintain documentation accuracy

Rationale: Strict hand hygiene is the single most effective measure to prevent hospital-acquired infections (HAIs) in the ICU, where patients are highly vulnerable.

QUICK REFERENCE — KEY NUMBERS & FACTS

Fact	Value / Detail
First modern ICU	1952, Copenhagen, polio epidemic
AACN founded	1969
Normal heart rate	60–100 bpm
GCS max/min	15 (normal) / 3 (deep coma)
GCS ≤ 8	Severe brain injury → consider intubation
Cushing's Triad	Hypertension + Bradycardia + Irregular respiration (raised ICP)
Pathological ICP	>20 mmHg
Head elevation for ICP	30° (reduces venous pressure in brain)
ICU nurse-patient ratio	1:1 or 1:2
AKI urine output criterion	<0.5 mL/kg/h for ≥6 hours
AKI creatinine criterion	↑ ≥0.3 mg/dL in 48h OR ≥1.5× baseline in 7 days
Sepsis bundle timing	Within 1 HOUR
Vasopressor for septic shock	Norepinephrine (first-line)
Sepsis fluid resuscitation	30 mL/kg crystalloid
HAI definition	≥48h after admission; not present at admission
ICU delirium tools	CAM-ICU, ICDSC
Enteral nutrition timing	Within 24–48h if feasible
Repositioning frequency	Every 2 hours
Refeeding syndrome	Hypophosphatemia + hypokalemia + hypomagnesemia
VAP prevention (key)	HOB 30-45° + daily oral care + daily sedation vacation
Most dangerous arrhythmia	Ventricular fibrillation → CPR + defibrillation
AFib main risk	Stroke (clot formation in atria)
Shock with slow pulse	Neurogenic shock (spinal injury)
Shock with warm early skin	Septic shock (early stage)
Emergency number (Cameroon)	112